Kennesaw State University 2026-2027 Undergraduate Catalog

Courses

Contract All Courses |

Computer Science

- ### CS 2290: Special Topics

3 Credit Hours

Prerequisite: Approval of the instructor, major area committee, and department chair.
Description
The course covers special topics at the intermediate level that are not in the regular course offerings.

- ### CS 3305: Data Structures

3 Credit Hours

Prerequisite: (MATH 2345 or CSE 2300) and ((CSE 1322 and CSE 1322L), or MTRE 2710 with a "B" or higher, or CPE 3000 with a "B" or higher)
Description
This course introduces data structures, specification, application, and implementation. The case studies will illustrate how data structures are used in computing applications. The emphasis of the course is on linear and some nonlinear data structures and object oriented principles. Topics include: abstract data types, stacks, queues, lists, binary search trees, priority queues, recursion, algorithm efficiency, trees, heaps, hash tables, and analysis of search and sort algorithms and their performance for implementation and manipulation. The programming language to be used in this course is any standard high-level object-oriented programming language such as C++, Java, and Ada.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Demonstrate understanding of single dimensional, multidimensional arrays, and dynamic arrays to store and access data, by applying these to programs.

2. Demonstrate learning of specifications and presentation of commonly used data structures by completely developing software. This must include appropriate documentation.
3. Demonstrate understanding and learn of advanced search and sort algorithms and their performance issues, by applying these to programs.
4. Demonstrate the analysis of the time complexity and space complexity of algorithms by explaining these concepts and applying these in programming.
5. Demonstrate the data structure covered in the courses by performing problem solving and application development.

- ## CS 3410: Introduction to Database Systems

3 Credit Hours

Prerequisite: CSE 1322 and CSE 1322L

Description

Introduction to database management systems, database processing, data modeling, database design, development, and implementation. Particular emphasis is placed on the relational approach to database management and processing, which focuses more on the logical nature of a database than its physical characteristics. Relational database programming assignments are drawn from the fields of business. Includes implementation of current DBMS tools and SQL. Ethical and security topics related to databases will be introduced.

- ## CS 3502: Operating Systems

3 Credit Hours

Prerequisite: CS 3305 and CS 3503

Description

The course covers the basic concepts, design and implementation of operating systems. Topics include an overview of basic computing hardware components, operating system structures, process management, memory management, file systems, input/output systems, protection and security. The Windows and/or UNIX/Linux operating systems will be reviewed as example systems.

- ## CS 3503: Computer Organization and Architecture

3 Credit Hours

Prerequisite: CSE 1322 and CSE 1322L

Description

The course covers computer architecture. Topics include data representation and encoding, binary arithmetic, fixed- and floating-point representation, numbering systems, error-control, instruction set architecture, assembly language and programming, microcode, memory organization and addressing, I/O, interrupts, internal architecture, instruction pipelining, multi-core architectures and CPU/ALU basics. The course also covers computer organization. Topics include Boolean Algebra,

logic gates, flip-flops, counters, registers, combinational and sequential circuits, K-maps, circuit design, and various digital components.

- ## CS 3622: Fundamentals of Data Communications

3 Credit Hours

Prerequisite: A grade of "B" or better in both CSE 1322 and CSE 1322L.

Description

An introduction on the fundamentals and underlying theory of data communication networks, their architecture, principles of operations and performance analyses. Topics include OSI reference model and standards, communication codes, network protocol concepts, synchronous and asynchronous transmission methods, line coding, signaling, effects of bandwidth and noise, digital and analog modulation, interfacing, error/flow/media-access control, switching and routing methods, and network topologies. Other areas studied are Local area networks (LANs), Wide area networks (WANs), Wireless networks, Fiber optic networks, internetworking technologies, and an introduction to the Internet, TCP/IP, cryptography and network security.

- ## CS 3626: Cryptography

3 Credit Hours

Prerequisite: MATH 2345 or CSE 2300

Concurrent: CS 3305

Description

The course covers both mathematical and practical foundations of cryptography. Topics include basic number theory for cryptography, conversion of text, and implementation using a programming language. The course includes historical cryptography, symmetric cryptography, asymmetric cryptography, hash functions, and well-known attack strategies with countermeasures. Exercises cover programming of simple cryptography in a programming language.

- ## CS 4265: Big Data Analytics

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)

Description

This course covers algorithms and tools that are needed to build MapReduce applications with Hadoop or Spark for processing gigabyte, terabyte, or petabyte-sized datasets on clusters of commodity hardware. A wide range of data algorithms will be discussed in this course.

- ## CS 4270: Intelligent Systems in Bioinformatics

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)

Description

Biological sciences are undergoing a revolution in how they are practiced. In the last decade, a vast amount of biological data has become available, and computational methods are playing a fundamental role in transforming this data into scientific understanding. Bioinformatics involves developing and applying computational methods for managing and analyzing information about the sequence, structure and function of biological molecules and systems. This course covers a wide range of machine learning, data mining, and computational algorithms to solve various bioinformatics research problems.

- ## CS 4306: Algorithm Analysis

3 Credit Hours

Prerequisite: CS 3305

Description

Algorithm analysis using formal and mathematical techniques and fundamental algorithm design strategies are studied. Topics include asymptotic analyses of complexity bounds using big-O, little-o, omega, and theta notations. The algorithmic strategies (brute-force, greedy, divide-and-conquer, recursive backtracking, dynamic programming, branch-and-bound, heuristics, and reduction) are covered. Also included are standard graph and tree algorithms. Additional topics include standard complexity classes, time-and-space tradeoffs in algorithms, and analyzing both recursive algorithms and non-recursive (iterative) algorithms.

- ## CS 4308: Concepts of Programming Languages

3 Credit Hours

Prerequisite: CS 3305 and CS 3503

Description

This course covers the fundamental concepts on which programming languages are based and the execution models supporting them. Topics include values, variables, bindings, type systems, control structures, exceptions, concurrency, and modularity. Languages representing different paradigms are introduced.

- ## CS 4322: Mobile Software Development

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153) and SWE 3313

Description

This course primarily focuses on mobile sensor application development and security of smartphones and mobile telecommunication systems. The goals of the course is to provide students with real world relevant mobile sensor app development and improve their knowledge and skills on mobile application development and mobile security.

- ## CS 4400: Directed Studies

1-3 Credit Hours

Prerequisite: Approval of the instructor, major area committee, and department chair.

Description

This course covers special topics of an advanced nature that are not in the regular course offerings. Up to three hours may be applied to the major area.

- ## CS 4412: Data Mining

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)

Description

This course covers fundamental data mining concepts and techniques for discovering interesting patterns from data in various applications. Topics include data preprocessing, data warehousing and OLAP, mining frequent patterns, classification, clustering, and tend analysis.

- ## CS 4422: Information Retrieval

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)

Description

This course covers the fundamentals of Internet search engines, including Web crawlers, inverted indices, hyperlink analysis, and relevance ranking. Also covered are advanced topics including information extraction for knowledge base construction, question answering, search marketing and ad targeting, and activity mining for relevance optimization and personalization.

- ## CS 4491: Advanced Topics in Computer Science

3 Credit Hours

Prerequisite: A grade of "C" or better in any prerequisite course. Prerequisite course(s) vary depending upon the topic.

Description

This course provides the current and relevant topics in an advanced Computer Science area of interest to faculty.

Notes: It may substitute for a CS major elective.

- ## CS 4492: Undergraduate Research

3 Credit Hours

Prerequisite: Department Permission and (CS 4265 or AI 4267 or CS 4270 or CS 4322 or CS 4412 or CS 4504 or CS 4512 or CS 4514 or CS 4522 or CS 4523 or CS 4524 or CS 4612 or CS 4622 or CS 4632 or CS 4712 or CS 4720 or CS 4722 or AI 4732).

Description

This course promotes undergraduate research in Computer Science. Students develop research ideas and conduct research work to investigate topics aligned with the department's research plan and with the guidance of Computer Science faculty members. Students document their findings in final reports, present their findings, and prepare research papers for publication in appropriate venues.

- ## CS 4493: Research Seminar

1 Credit Hours

Prerequisite: Department Permission and GPA of 3.0 or higher.

Description

Discussion of the latest developments and research areas in Computer Science. Students will share among each other research ideas related to current research areas with interactions with faculty or invited speakers. Students will compile, discuss, present, and report their research ideas.

- ## CS 4504: Parallel and Distributed Computing

3 Credit Hours

Prerequisite: CS 3305 and CS 3503

Concurrent: CS 3502

Description

This course introduces students to the fundamental principles common to the design and implementation of programs that run on two or more interconnected computer systems - in parallel or distributed configurations. Topics to be covered include: essentials of operating systems, network protocols for process communication, and synchronization using message queues; understanding of client-server paradigms, web-based group or collaborative communication systems; advanced distributed computing paradigms for parallel computing and handling concurrency issues; and sockets. Programming will focus on using API's for parallel or distributed applications (e.g., MPI and RMI).

- ## CS 4512: Systems Programming

3 Credit Hours

Prerequisite: CS 3305 and CS 3502

Description

This course presents an introduction to systems programming in Linux/Unix. Topics include file I/O, process control and communication, threading, and network-aware systems programs.

- ## CS 4514: Real-Time Systems

3 Credit Hours

Prerequisite: CS 3502

Description

This course covers the software-development life cycle as it applies to real-time systems. Alternatives: • Including labs that involve the use of a real-time operating system and an associated development environment, or • Modeling with UML, and object oriented simulation. Introduction to formal specification of real-time systems. A course project is required to be completed by the end of the semester.

- ## CS 4522: HPC & Parallel Programming

3 Credit Hours

Prerequisite: CS 4504

Description

This course will introduce parallel programming techniques for shared memory and distributed memory systems. Topics include threading, OpenMP, and MPI.

- ## CS 4523: Programming Massively Parallel Processors

3 Credit Hours

Prerequisite: CS 3305 and CS 3502

Description

A study of practical parallel algorithms with an emphasis on implementation and performance issues on massively parallel processors. Design and implement high performance computing applications using CUDA running on Graphics Processing Unit (GPU). Topics include heterogeneous parallel programming, hardware threading models, synchronization, parallel blocking algorithms, register allocations, memory performance, and inter-thread communication.

- ## CS 4524: Cloud Computing

3 Credit Hours

Prerequisite: CS 4504

Description

This course discusses the fundamental concepts and techniques of cloud computing. Students will develop an understanding of cloud computing architecture, Infrastructure as a Service (IaaS), Platform-as-a-Service (PaaS), Software as a Service (SaaS), Virtualization, and Application Development on Cloud.

- ## CS 4612: Software Security

3 Credit Hours

Prerequisite: CS 3502 and CS 3626

Description

The course introduces the fundamental concepts and principles of software security. Topics covered include buffer overflows, defense mechanisms, return oriented programming, reverse engineering, vulnerabilities analysis. Additional topics: mobile security, hardware platform security, embedded system security.

- ## CS 4622: Computer Networks

3 Credit Hours

Prerequisite: CS 3503 and CS 3622

Description

This course covers computer networking and includes software application-related, protocol-related and security-related issues involved in the Internet. Topics include basic network structures, mechanisms for application-to-application communications, protocol layering, Internet addressing, unicast and multicast routing, connection establishment and termination, data flow and congestion control, and error handling. A specific protocol suite will be examined in detail. More advanced

topics that build on the student's understanding of network protocols are also introduced, such as network security, mobile networks and the future Internet.

- ## CS 4626: Computer and Network Security

3 Credit Hours

Prerequisite: CS 3626 and CS 4622

Description

This course is a comprehensive study of the security principles and practices for computer systems and networks. Topics to be covered include basic security concepts, common network attack techniques and impacts, common security policies, Internet protocol vulnerabilities, basic cryptographic tools and secure protocols. Defense techniques such as authentication, access control, encryption, and network intrusion detection will be discussed. This course will also survey new and emerging network security topics, applications and technologies. Exercises for this course will include network programming in some language and using various tools in understanding and analyzing packet traces and network traffic.

- ## CS 4632: Modeling and Simulation

3 Credit Hours

Prerequisite: CS 3305

Description

This course covers the modeling and simulation of the structure and behavior of real-world systems using object-oriented discrete-event simulation techniques. The course emphasizes the modeling and computer programming perspective of simulation; design and implementation of simulation models. The fundamental concepts of object-oriented simulation are introduced. Model implementation will require programming in an object-oriented simulation language such as OOSimL, or in a general purpose programming language (Java or C++). Students will also be exposed to a commercial integrated simulation software tool: Arena.

- ## CS 4712: User Interface Engineering

3 Credit Hours

Prerequisite: CSE 1322 and CSE 1322L

Description

A comprehensive study of techniques in design and implementation of user interfaces engineering. Topics include the foundation of human-computer interaction and interface related to software lifecycle, building a graphic user interface engineering, interaction devices and technologies, human-computer dialogue, cognitive models, usability, the design and development process, user interface management systems (UIMS), interface style and techniques, user learning, and diversity in interaction styles. Major research and the building of a working graphic user interface are included.

- ## CS 4720: Internet Programming

3 Credit Hours

Prerequisite: CS 3305 and (CS 3410 or CSE 3153)

Description

This course introduces current technologies for modeling, designing, implementing, and developing Web applications. Topics include developing for the server and the client, programming frameworks, server administration and integration with databases. Practice will involve platforms and language such as Linux, Python, PHP, Ruby and JavaScript.

- ## CS 4722: Computer Graphics and Multimedia

3 Credit Hours

Prerequisite: CS 3305

Description

The basic principles and practices of interactive computer graphics and multimedia systems are covered in this introductory course. The design and implementation of state-of-the-art computer graphic rendering and visual multimedia systems are the main part of the course. The sub-topics of the course deal with specific input/output hardware devices and their technology, software and hardware standards, programming methods for implementing 3-dimensional graphical applications and interactive multimedia applications, and a study and evaluation of the effectiveness of graphic/multimedia communications. A large component of the class is the building of a large-scale application.

- ## CS 4850: Computer Science Senior Project

3 Credit Hours

Prerequisite: CS 3502 and SWE 3313

Description

This course provides a capstone experience for CS majors to promote a successful transition into the work place or further academic study. Students will have the opportunity to practice essential project management skills and work with current software tools and technologies. Student teams will develop a project scope, project plan, document functional specifications, develop a design document, implement specified functions, provide weekly progress reports, give project presentations to the class, conduct final project presentation to the instructor and/or project sponsor, and provide a complete final report that includes documentation of all class activities. Each team will designate a team leader who is responsible for coordinating work tasks, team meetings, communications with the instructor and/or project sponsor, and team effort.

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